RAJALAKSHMI ENGINEERING COLLEGE

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A2

Started on	Wednesday, 19 August 2026, 8:26 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:31 AM
Time taken	5 mins 38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the index of the first zero using binary search
int findFirstZero(int arr[], int low, int high, int size) {
    if (high >= low) {
        int mid = low + (high - low) / 2;

        // Check if mid is the first zero
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }
        // If the element at mid is 1, search in the right half
        else if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high, size);
        }
        // Else search in the left half
        else {
            return findFirstZero(arr, low, mid - 1, size);
        }
    }
    return -1; // No zero found
}

int main() {
    int m;

    // Read size of array
    if (scanf("%d", &m) != 1 || m <= 0) {
        printf("Invalid array size.\n");
        return 1;
    }

    int *arr = (int *)malloc(m * sizeof(int));
    if (!arr) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    // Read array elements
    for (int i = 0; i < m; i++) {
        if (scanf("%d", &arr[i]) != 1 || (arr[i] != 0 && arr[i] != 1)) {
            printf("Invalid input. Only 0s and 1s allowed.\n");
            free(arr);
            return 1;
        }
    }

    // Find the first zero index
    int firstZeroIndex = findFirstZero(arr, 0, m - 1, m);

    // Calculate number of zeros
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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